

Fig. 1.14

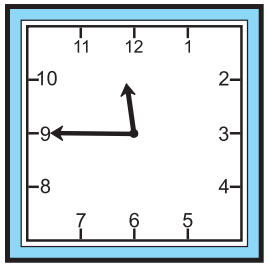


Fig. 1.15

At quarter past 5, miniute hand is pointing to 3. Hour hand has gone past 5. So the angle between them is more than 60º. In one minute hour hand turns through $\frac{1}{2}^{o}$ hence in 15 minutes it has turned through $\left(\frac{15}{2}\right)^{o} = 7.5º$.

Thus the angle between the hands is equal to 60º + 7.5º = 67.5º.

ii) At quarter to twelve, minute hand is pointing to 9, hour hand is between 11 and 12 though it is nearer 12. It will take 15 minutes i.e. 7.5º to reach 12.

∴ the angle between the hands is equal to 90º – 7.5º = 82.5º.

Note:

In degrees	0°	30°	45°	60°	90°	180°	270°	360
In radians	0^c	$\frac{\pi^c}{6}$	$\frac{\pi^c}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π^c	$\frac{\pi^c}{4}$	$2\pi^c$

EXERCISE 1.1

Q.1 A) Determine which of the following pairs of angles are co-terminal.

i) 210°, –150° ii) 360°, –30°

iii) –180°, 540° iv) –405°, 675°

v) 860°, 580° vi) 900°, –900°

B) Draw the angles of the following measures and determine their quadrants.

i) –140° ii) 250° iii) 420° iv) 750°

v) 945° vi) 1120° vii) –80° viii) –330°

ix) –500° x) –820°

Q.2 Convert the following angles in to radian.

i) 85° ii) 250°

iii) –132° iv) 65°30′

v) 75°30′ vi) 40°48′

Q.3 Convert the following angles in degree.

i) $\frac{7\pi^c}{12}$ ii) $\frac{-5\pi^c}{3}$ iii) 5^c

iv) $\frac{11\pi^c}{18}$ v) $\left(\frac{-1}{4}\right)^c$

Q.4 Express the following angles in degree, minute and second.

i) (183.7)° ii) $(245.33)^0$ iii) $\left(\frac{1}{5}\right)^c$

Q.5 In Δ ABC, if $m\angle A = \frac{7\pi^c}{36}$, $m\angle B = 120°$, find $m\angle C$ in degree and radian.

Q.6 Two angles of a triangle are $\frac{5\pi^c}{9}$ and $\frac{5\pi^c}{18}$. Find the degree and radian measure of third angle.

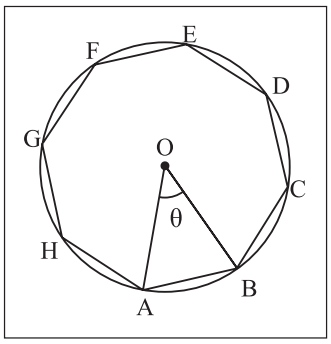
Q.7 In a right angled triangle, the acute angles are in the ratio 4:5. Find the angles of the triangle in degree and radian.

Q.8 The sum of two angles is $5\pi^c$ and their difference is 60°. Find their measures in degree.

Q.9 The measures of the angles of a triangle are in the ratio 3:7:8. Find their measures in degree and radian.

Solution : Here r = 9cm

$\theta = \frac{360^\circ}{8} = 45^\circ = \frac{\pi^c}{4}$



l(minor arc AB) = S

$= r\theta$

$= 9\left(\frac{\pi}{4}\right)$ cm

Fig. 1.20

EXERCISE 1.2

(1) Find the length of an arc of a circle which subtends an angle of 108° at the centre, if the radius of the circle is 15 cm.

(2) The radius of a circle is 9 cm. Find the length of an arc of this circle which cuts off a chord of length, equal to length of radius.

(3) Find the angle in degree subtended at the centre of a circle by an arc whose length is 15 cm, if the radius of the circle is 25 cm.

(4) A pendulum of length 14 cm oscillates through an angle of 18°. Find the length of its path.

(5) Two arcs of the same lengths subtend angles of 60° and 75° at the centres of two circles. What is the ratio of radii of two circles ?

(6) The area of a circle is 25π sq.cm. Find the length of its arc subtending an angle of 144° at the centre. Also find the area of the corresponding sector.

(7) OAB is a sector of the circle having centre at O and radius 12 cm. If m∠AOB = 45°, find the difference between the area of sector OAB and triangle AOB.

(8) OPQ is the sector of a circle having centre at O and radius 15 cm. If m∠POQ = 30°, find the area enclosed by arc PQ and chord PQ.

(9) The perimeter of a sector of the circle of area 25π sq.cm is 20 cm. Find the area of the sector.

(10) The perimeter of the sector of the circle of area 64π sq.cm is 56 cm. Find the area of the sector.



Let's Remember

- If an angle is r radians and also θ degrees then $\frac{r}{\pi} = \frac{\theta}{180^\circ}$
- $\theta^\circ = \left(\theta \times \frac{\pi}{180}\right)^c$, $1^\circ = (0.01745)^c$
- $\theta^c = \left(\theta \times \frac{180}{\pi}\right)^\circ$, $1^c = 57^\circ 17' 48''$
- Arc length = s = $r\theta$. θ is in radians.
- Area of a sector A = $\frac{1}{2}r^2\theta$, where θ is in radians.
- Two angles are co-terminal if and only if the difference of their measures is an integral multiple of 360.
- Exterior angle of a regular polygon of n sides = $\left(\frac{360}{n}\right)^\circ$
- In one hour, hour's hand covers 30° and a minutes hand covers 360°.

- In 1 minute, hour hand turns through $\left(\frac{1}{2}\right)^{o}$ and minute hand turns through 6°.

MISCELLANEOUS EXERCISE - 1

I Select the correct option from the given alternatives.

1) $\left(\frac{22\pi}{15}\right)^{c}$ is equal to

A) 246° B) 264° C) 224° D) 426°

2) 156° is equal to

A) $\left(\frac{17\pi}{15}\right)^{c}$ B) $\left(\frac{13\pi}{15}\right)^{c}$ C) $\left(\frac{11\pi}{15}\right)^{c}$ D) $\left(\frac{7\pi}{15}\right)^{c}$

3) A horse is tied to a post by a rope. If the horse moves along a circular path, always keeping the rope tight and describes 88 meters when it traces the angle of 72° at the centre, then the length of the rope is

A) 70 m. B) 55 m. C) 40 m. D) 35 m.

4) If a 14cm long pendulum oscillates through an angle of 12°, then find the length of its path.

A) $\frac{13\pi}{14}$ B) $\frac{14\pi}{13}$ C) $\frac{15\pi}{14}$ D) $\frac{14\pi}{15}$

5) Angle between hands of a clock when it shows the time 9.45 is

A) $(7.5)^{o}$ B) $(12.5)^{o}$ C) $(17.5)^{o}$ D) $(22.5)^{o}$

6) 20 meters of wire is available for fancing off a flower-bed in the form of a circular sector of radius 5 meters, then the maximum area (in sq. m.) of the flower-bed is

A) 15 B) 20 C) 25 D) 30

7) If the angles of a triangle are in the ratio 1:2:3, then the smallest angle in radian is

A) $\frac{\pi}{3}$ B) $\frac{\pi}{6}$ C) $\frac{\pi}{2}$ D) $\frac{\pi}{9}$

8) A semicircle is divided into two sectors whose angles are in the ratio 4:5. Find the ratio of their areas?

A) 5:1 B) 4:5 C) 5:4 D) 3:4

9) Find the measure of the angle between hour-hand and the minute hand of a clock at twenty minutes past two.

A) 50° B) 60° C) 54° D) 65°

10) The central angle of a sector of circle of area 9π sq.cm is 60^{0}, the perimeter of the sector is

A) π B) 3+π C) 6+π D) 6

II Answer the following.

1) Find the number of sides of a regular polygon if each of its interior angle is $\frac{3\pi^{c}}{4}$.

2) Two circles, each of radius 7 cm, intersect each other. The distance between their centres is $7\sqrt{2}$ cm. Find the area of the portion common to both the circles.

3) Δ PQR is an equilateral triangle with side 18 cm. A circle is drawn on the segment QR as diameter. Find the length of the arc of this circle within the triangle.

4) Find the radius of the circle in which central angle of 60° intercepts an arc of length 37.4 cm.